SDS 210 - Introduction to Statistics

Katherine M. Kinnaird

1. Welcome to SDS 220

I am so excited that you'll be joining this educational journey through statistics! You are now, just by being here, a statistician. This course will help us frame out exactly what that means and also what *statistics* is.

> **Note**
>
> You are now a statistician!

1.-.i. *Official Course Description:*

An application-oriented introduction to modern statistical inference: study design, descriptive statistics; random variables; probability and sampling distributions; point and interval estimates; hypothesis tests, resampling procedures and multiple regression. A wide variety of applications from the natural and social sciences are used. Classes meet for lecture/discussion and for a required laboratory that emphasizes analysis of real data. SDS 220 satisfies the basic requirement for biological science, engineering, environmental science, neuroscience and psychology.

Motivating Questions:

There are a few questions motivating our course work:

- What is Statistics?
- What role does data play in statistics?
- What habits of mind do we need to develop to be effective and ethical statisticians?

Course Learning Objectives:

By the end of the course, statisticians (ie. you!) will be able to…

1. Responsibly use sample data to learn about a larger population, using exploratory and inferential statistics - *Create Statistics*
2. Detail the power and limitations of statistical concepts, and use this understanding to evaluate statistics from various contexts - *Evaluate Statistics*
3. Explain the connection between probability and inferential statistics - *Connect Statistical Concepts*
4. Work collaboratively, responsibly, and reflectively to create and present statistics to a wide variety of audiences - *Communicate with Statistics*

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1.a. Syllabus

This webpage and the associated links together function as the syllabus. Everything that you need to know about the course is either here or linked from here.

The course syllabus is the most important document underlying the culture of our course and our classroom. I view my syllabus as a sacred document that both introduces and governs the course. In taking this view, I work to detail as much about the course as possible from the big picture ideas to the minute details of course policies. I believe that the first activity a student should do when beginning a course is to carefully read and examine the syllabus. In keeping with that belief, there are a few tasks in this document that will help us build our course community.

> **Note**
>
> Please read the syllabus this week and complete these tasks to help shape our community. If you have questions about the syllabus, please ask them!

1.b. Course Acknowledgements

Parts of this course - including slides, activities, and notes - will be from a variety of sources. Materials will be appropriately attributed and will be used in keeping with copyright and fair use laws.

2. Course Philosophy

This course is designed to be a first course in Statistics for students with **no prior experience** in or with statistics. This course seeks to begin your training in statistics. Success in this course will be measured iteratively, placing emphasis on the student's ownership of the learning process and a student's consistency of effort.

2.a. Stretch Zone

Borrowing language from the 2019 Google J-term course, when learning in a classroom setting, at any moment, we exist in one of three zones: safe, stretch, or strain. A student in the safe zone is completely comfortable, and the course material is completely within what the student already knows. The strain zone is the complete opposite of the safe zone. A student in the strain zone feels completely overwhelmed with the material to the point where they can't find any meaningful connections between the current material and what the student already knows.

The material in this course is intellectually challenging. Several of the topics are ones that took me a long time to fully understand their subtleties. The goal for this course is to spend the vast majority of our time in the stretch zone. In this zone, the material is just beyond our current knowledge base and while we are a bit uncomfortable with the material, we can see possible paths to tie the new material into our existing knowledge. In other words, in this class, our brains should be stretching, but not straining.

At the end of each week, if you have not been surprised or challenged, or if you have no questions about the material, this could mean that you are in the safe or strain zones. If you find yourself exasperated or negative about the class, you may be in the strain zone. If you find yourself using words like "boring" or "easy", you are likely in the safe zone. If you are in either zone, please get in touch with me: in class, on slack, in an appointment, or send me an email. I want you to get the most out of this course, but I cannot make adjustments to the course if I do not have firsthand knowledge of what is going on.

3. Professional Development

As we explore statistics, we will also be developing as professional statisticians. Specifically, we will focus on what it means to be a member of a professional environment and how to be intentional and reflective in our work.

3.a. Participating in the Course

Being part of a professional community has many parts; we generate and consume knowledge, we review and critique work, and we present to and learn from each other. The edge of what we know is constantly being pushed further and further as we discover new ideas, unearth new techniques, and ask new questions. At times, we are programmers, writers, and gardeners, and then at other times, we serve as reviewers, mentors, and learners. The constancy is our willingness to be present and active within our professional community. With this in mind, in this course, your intentional participation is **expected.**

Participation can take many forms, including but not limited to: attending class meetings, being active on our course slack, participating in discussion in class, offering ideas and questions in class or on slack. As a community our class will generate a list on our google drive of what we believe are ways to engage and participate in our course.

> **Note**
>
> Our class's definition for participation can be found here

Adding to this list one way that you are called to participate in this course. If you find yourself unwilling or resistant to add to this list, examine this. What is holding you back? What would help propel you to action?

3.b. Preparing for Class

Participating our course will require more than being physically present; it is being prepared to engage with the topics, concepts, and activities of the day. This means having completed the reading, and any other pre-class activities for the day. *Completion* in this context means giving yourself the time and space to finish the most recent reading (asking questions on slack as you have them), to formulate half-formed questions and comments about the readings, and to try, fail, and try again on the group problems. In this way, completion is more about expanding our ideas (by coming up with questions and comments) about machine learning, than it is about perfectly finishing all the things. This kind of preparation means:

- Blocking time in your daily work life for this course
- Noting and sharing each question you have
- Reading with a pen and highlighter in your hand
- Taking frequent notes on readings, discussions, and labs
- Working on pieces of the group problems and reviewing what your group discussed

- Gathering the fragments of answers to the questions that you and others have posed

3.b.i. *Attending class:*

We are still in the midst of a global pandemic. In keeping with the Guidelines laid out in Smith's Culture of Care, if you are ill and/or have *any* COVID symptoms, please do **not** come to *in-person* class. Instead, please log in on zoom. The link can be found on our Moodle site and our slack space.

You do not need to email me to ask permission to come to class over zoom or in person. However, if you are not able to be in-person for 3 consecutive meetings, then we need to check in.

Similarly, do not come to student hours nor appointments if you are ill and/or have *any* COVID symptoms. There will be a zoom link for student hours and a zoom link can be provided for any in-person appointment.

> **Warning**
>
> Do **NOT** come to in-person class or student hours, if you are ill and/or have any COVID symptoms.

Failure to respect this policy will result in an email to both the class dean and your advisor.

3.c. Intentional and Reflective Work

A project involving statistical procedures is rarely a straightforward pursuit with a clear path from beginning to end. I have found the most success when I engage a process of constantly reflecting on where I am in a project compared to where I began and then intentionally determining my next step based on those assessments. In this class, we will reflect frequently. First we will mark the beginning and the end of the semester with a general check-in on where we are in our educational journeys. Throughout the course, you will reflect on what you learned in each lab and will have space to ask questions that are still lingering for you.

3.d. Writing

This course focuses on statistics and while it might be tempting to assume that "statistics" means "complicated and inaccessible writing", this is not true for our course. For each assignment, you will be asked to explain your process, detail how the various pieces interact, and illuminate any potential weaknesses. This is excellent training for the 'real world' where you will have managers, users, and customers with varying levels of technical expertise.

In your assignments, I expect that you treat your readers as colleagues who have your respect and who you are genuinely interested in sharing your ideas with. This means that you may need to edit your responses or rewrite a solution to a problem for additional clarity. If you find yourself writing "the thing" or "you know what I mean," I would recommend taking that as a sign that your assignment needs additional editing. If you have any questions about writing for statistics or more general, you are always welcome to reach out to me.

3.d.i. *Writing Enriched Curriculum:*

The project will also scaffold explicit development of writing skills that SDS has laid out in our department's newly adopted writing plan. Specifically, our course will focus on three of the listed writing goals:

- Writing Goal 2 - Engage in a writing process that includes brainstorming, outlining, initial drafting, peer review, editing, and revising;
- Writing Goal 4 - Prioritize the important parts of the process and/or project to communicate
- Writing Goal 5 - Clearly communicating the research question and how analysis will support the question in the introduction.

3.d.ii. *Jacobson Center for Writing, Teaching & Learning:*

Smith has an additional resource for writing support: the Jacobson Center for Writing, Teaching & Learning make an appointment to take your work to the Jacobson Center on their website. In particular, you may choose to bring your work to Peer Writing Tutors Elisabeth Nesmith or Elina Gordon-Halpern, both SDS majors who tutor for the Jacobson Center. Contact Sara Eddy (seddy@smith.edu) for more information about their schedules or how to make an appointment.

The above language about the Jacobson Center was adapted from language from Sara Eddy. Used with permission

4. Course Logistics

4.a. Class Meeting Times

The course has two parts: Lecture and Lab. The lecture meets Mondays, Wednesdays and Fridays in Sabin-Reed 301 from 10:50am to 12:05pm. Half of the class will be in Lab 1 which meets Thursdays from 9:25am to 10:40am also in Sabin-Reed 301. Another half of the class will be in Lab 2 which also meets Thursdays in Sabin-Reed 301, but from 10:50am to 12:05pm. In supporting our course community, students should make every effort to be at our lecture and lab meetings on time.

> **Note**
>
> Please attend the lab that you are registered for!

4.a.i. *Attending Lecture and Lab:*

We are still in the midst of a global pandemic. In keeping with the Guidelines laid out in Smith's Culture of Care, if you are ill and/or have *any* COVID symptoms, please do **not** come to *in-person* class. Instead, please log in on zoom. The link can be found on our Moodle site and our slack space.

You do not need to email me to ask permission to come to class over zoom or in person. However, if you are not able to be in-person for 3 consecutive meetings, then we need to check in.

Similarly, do not come to student hours nor appointments if you are ill and/or have *any* COVID symptoms. There will be a zoom link for student hours and a zoom link can be provided for any in-person appointment.

> **Warning**
>
> Do **NOT** come to in-person class or student hours, if you are ill and/or have any COVID symptoms.

Failure to respect this policy will result in an email to both the class dean and your advisor.

4.a.ii. *Pivoting Class Meetings:*

If I am unwell and/or experiencing any symptoms of COVID, class will be held on zoom. If this happens, I will send a slack message on the `#general` channel using the `@everyone` mention. Using `@everyone` should send an email to your inbox.

4.b. Communication

In addition to our synchronous meetings, our class will make of electronic communication, including our class slack, email, and Moodle messages. These methods of communication represent differing levels of formality and collaboration:

- **Slack:** Our slack site is the primary form of course communication. It allows for us to share where we are stuck with a reading, idea, or assignment, and where we can add helpful hints, request study groups, or share interesting news articles. Slack is much less formal than email, and salutations and signatures are not required. If you need to ask your instructor a question, this is the best place to do it either over a public channel or through direct message.
- **Email:** Email communication is more formal than slack. It should be used in the cases that 1) concern something personal, 2) require attachments, or 3) involve a number of people (who should be copied on the email). We also will use email to confirm individual appointments. Emails to the instructor should include a salutation and a signature.
- **Slack Mentions:** When I need to communicate with the whole class on a time-sensitive manner, I will use the `@everybody` or `@channel` functionality in slack. Please check that your slack settings will notify you either via slack or email. If you are using the email notifications, please be sure that your email does not treat these messages as spam.
- **Slack Emojis:** At times, I will ask you to emoji a message to show that you have read the message. This is a quick way for you to signal that you read it. If there's something that you have a question about in a "please emoji" message, please use the threading function for that message or send me a direct message on slack.

> **Note**
>
> Please sign up for our slack site, add a profile picture, and introduce yourself with your name and a fun-fact about yourself on the #intros channel. After signing up, please determine how and where you will get notifications from slack.

Please note that on a typical workday, I leave my office just after 3:30pm. This means that emails and slack messages sent after 3:30pm will likely be received on the next business day. Similarly e-communication sent on the weekend will likely not be received until Monday.

4.c. Book

Our book is *Introduction to Modern Statistics* by Mine Çetinkaya-Rundel and Johanna Hardin. It is part of the OpenIntro project and as such is freely available as a PDF or on the web. Printed copies can be purchased for $20 at the Smith College Bookstore or from Amazon.

> **Note**
>
> The web-native version of our book *Introduction to Modern Statistics* can be found here

4.d. Instructor Information

The instructor for this course is Katherine Kinnaird. My office is room 208 in McConnell Hall. My email is kkinnaird@smith.edu. My student hours are Mondays from 1:30pm to 3:30pm, Fridays from 9am to 10am, and by appointment.

> **Note**
>
> To book an appointment with me, go to my appointment calendar

5. Assignments

To prepare students to tackle the course's motivating questions - especially *What habits of mind do we need to develop to be effective and ethical statisticians?* - there are a variety of assignments in this course. Each week will be a mix of reading about topics in statistics, discussing and debating concepts, putting our readings into practice, and constant reflection about our learning and professional development. Due dates are posted on the Detailed Course Schedule and are due on the stated dates at 10am.

> **Note**
>
> The Detailed Course Schedule lists the due dates for the course. To view it, please log into your **Smith** account.

5.a. Engagement & Class Preparation

For each lecture meeting and lab, there will be readings and/or a pre-class activity. These readings and activities are intended to be your first contact with the course material. Engaging with these materials in good faith *before* class will help you get the most out of the in-class activities.

5.b. Group Problem Sets (GPsets)

Group Problem Sets are designed to be completed in class with possibly some work out of class to clarify details that your group discussed. The goal of these GPsets is to put the ideas from the readings and pencasts to work in examples.

GPsets are due on **Thursdays at 10am** unless otherwise noted on the Detailed Course Schedule. Often there will be more than one GPset due on each Thursday. Please consult the Detailed Course Schedule.

5.c. Labs

The labs are designed to be completed in class with some work out of class to practice the ideas from the pre-lab assignments. Pre-labs are due **before the start of lab on Thursdays**. Labs are due on **Fridays at 10am** unless otherwise noted on the Detailed Course Schedule.

5.d. Projects

There will be one group project for the semester. In this project, you will use what we have learned in this class to investigate one data set of your choosing. There will be lab time set aside to work on the project.

The project will also scaffold explicit development of writing skills that SDS has laid out in our writing plan. Specifically, our course will focus on three of the listed writing goals:

- Writing Goal 2 - Engage in a writing process that includes brainstorming, outlining, initial drafting, peer review, editing, and revising;
- Writing Goal 4 - Prioritize the important parts of the process and/or project to communicate

- Writing Goal 5 - Clearly communicating the research question and how analysis will support the question in the introduction.

5.e. Starred Problems

Each week, you will complete a starred problem. The goals of these problems are to 1) apply what we've discussed in class on your own, 2) reflect on your learning so far, and 3) practice for the knowledge checks. To accomplish these goals, starred problems will all follow the same procedure:

1. Study what we have learned so far
2. Put away all course resources
3. Download that week's starred problem off Moodle
4. Write your solution to the problem and upload it into Moodle
5. Once properly uploaded, the instructor solution will appear
6. Compare your solution to the instructor solution 6.Complete a reflection worksheet about this problem and what you would like to review as a result of doing this problem

These problems are based on completion (ie. you get the credit or not) based on both your solution and the content of your reflection. This means that you can get the problem totally wrong and still get full credit if you take the time to honestly reflect on your progress. This assignment structure is based on an assignment structure from Smith's Engineering Program.

Starred problems are due on **Mondays at 10am** unless otherwise noted on the Detailed Course Schedule.

5.f. Knowledge Checks

There are 4 Knowledge Checks, during which you can collect statistics medals. There are 15 medals, each representing core skills for the course. During each knowledge check, you can attempt as many or as few medals as you would like. Your goal, by the end of the semester, is to collect all 15 medals.

The first badge challenge will have the first 5 medals available, the second will have the first 10, and the third and fourth knowledge check will have all 15 medals. You do not need to (nor should you) attempt all medals on each knowledge check, but you do need to collect all 15 medals by the end of the term.

> **Note**
>
> The weekly starred problems will be in the style of the knowledge check questions.

5.f.i. *Statistics Medals:*

The 15 medals that represent core skills for the course are listed below, in the order that they appear in the course:

1. Describe and evaluate context for a statistic, statistical test, or procedure, including any sampling biases and the data collection design
2. Compute and contextualize notions of typical and spread
3. Create appropriate graphical summaries given the data and context of the data; Articulate the limitations and strengths of each type of graphical summary
4. Compare and contrast sample distributions and sampling distributions; Compare and contrast sampling and resampling; Compute and accurately describe standard error to a non-expert
5. Compute and accurately describe confidence intervals to a non-expert; Articulate how confidence intervals relate to statistical inference
6. Compute appropriate probabilities within a given context; State and apply Bayes' Rule
7. State the four steps of a hypothesis test; Relate hypothesis testing to probability distributions; Detail the limits and potential abuses of hypothesis testing
8. Define Type I and Type II errors; Explain how Type I and Type II are connected
9. Describe issues of running multiple hypotheses in one hypothesis test to both an expert (using probabilistic ideas for support) and a non-expert
10. Create linear models; Detail the meaning of each coefficient in a contextually meaningful fashion
11. Evaluate linear models using statistical measures; Articulate limits and potential abuses of using linear models
12. Identify potential confounding variables; Identify if, where, and how Simpson's paradox is at play in a given context
13. State and apply the Central Limit Theorem; Describe how the Central Limit Theorem relates to the normal distribution
14. Describe the normal distribution including detailing features related to skew and standard deviation; articulate the differences between normal and t-distributions; articulate the differences between cumulative and density curves
15. Apply ANOVA and χ^2 tests appropriately, articulating each step of the tests

5.g. Weekly flow for the course

Each week, you will have group problem sets, a lab and either a starred problem or a knowledge check due each week. The course will operate as follows:

Day	Course Preparation and/or Activity	Assignment due
Monday	Group Problem Set in class	Starred problem
Wednesday	Group Problem Set in class	(none)
Thursday	Lab in Class	Pre-lab; Group Problem sets from Monday, Wednesday, and previous Friday

Day	Course Preparation and/or Activity	Assignment due
Friday	Group Problem Set in class	Lab from Thursday

5.h. Workload

According to federal standards, each five-credit course should equate to at least 225 hours of work over the semester. If you are taking 16 credits, that equates to 720 hours of work over the 15 weeks of the semester, from the first day of classes until the end of the final exam period. In the case of this course, you will spend nearly 15 hours each week on my class alone (including our six hours of class meetings per week).[1]

In considering the work for this course, I believe that the approximate 9 hours per week *outside* of class will breakdown something along these lines:

- 3 hours of class prep - One hour per lecture for reading the book and prep activities
- 1 hour per week of lab prep and/or wrap up
- 1 hour per week on the project
- 1 hour per week on the starred problem
- 3 hours of "flex time" used to supplement any of the above areas and/or go to student hours

Notice that there are 3 hours of flex time. This is to accommodate weeks were you might want to spend more time on an assignment. For example, you might spend extra time studying in the weeks leading up to a knowledge check.

If you find that the time you are spending on this class is a lot more than 9 hours per week *outside* of class or a lot less, let's check in.

5.h.i. *Spending time on* You*:*

The current system is designed to keep you working more than 40 hours per week just on your courses. This means that the hours not working on class work are precious and should be treated as such. This does not mean that every hour of every day should be filled with things deemed important by others, but more as a call to be intentional with your time. A few questions you might want to ask yourself include:

- Are you spending time on yourself and your wellbeing? What are you doing to restore yourself? How much time and space do these activities need to be effective?
- Are you giving to your courses the time that you want to be? Are you giving them the time during part of the day when your brain is ready to engage new ideas?

[1] Each of our 75 minute meetings is "counted" as 90 minutes. Since we meet 4 times per week, our "in-class contact time" is counted as 6 hours.

- Are you working doing your best thinking hours? If so, do you have any flexibility to adjust your daily schedule to be more present with your studies?
- Are you devoting time to being with your friends and/or your family? Are you present during those times?
- Are you losing hours to things or activities that are not high on your priority list? Alternatively, are you losing hours or days that you cannot recall what you did that day or why it so consumed you?

The adage tells us that “time is our most precious resource.” This statement frustrated me as a student because “I’ll have time after college/graduate school/post doc.” I have come to believe that this moment is the time that we have and if we are not intentional with our time today, right now, we will lose, not hours, not days, but months and years. I am working on a process of aligning my time to reflect my priorities as an educator, a researcher, a friend, and a family member. This is not a process that is completed in a few hours or a few days, but rather it is an iterative process of repeated and constant reflection.

6. Resources Supporting SDS 220

6.a. Inclusivity for all students

Smith is committed to providing support services and reasonable accommodations to all students with disabilities. Please inform me early in the term if there are aspects of the course that need to be modified to serve your learning, health, or person. You may speak with me after class, during student hours, or during an appointment. To request an accommodation, please register with the Disability Services Office at the beginning of the semester. To begin the process, either call 413-585-2071 or email ods@smith.edu.

Students in need of short-term academic advice or support can contact your class dean in the Dean of the College office. If you require some accommodation for religious or cultural purposes, please do not hesitate to talk to me about this.

6.b. Spinelli Center

The Spinelli Center for Quantitative Learning is a resource to support students as they take STEM courses and as they engage in other quantitative work. To see the list of available peer tutors and other resources available at the Spinelli Center, please consult their website.

6.c. Jacobson Center

Smith has an additional resource for writing support: the Jacobson Center for Writing, Teaching & Learning make an appointment to take your work to the Jacobson Center on their website. In particular, you may choose to bring your work to Peer Writing Tutors Elisabeth Nesmith or Elina Gordon-Halpern, both SDS majors who tutor for the Jacobson Center. Contact Sara Eddy (seddy@smith.edu) for more information about their schedules or how to make an appointment.

The above language about the Jacobson Center was adapted from language from Sara Eddy. Used with permission

6.d. Library

Our book is freely available on the web as a part of the OpenIntro project.

> **Note**
>
> The web-native version of our book *Introduction to Modern Statistics* can be found here

The library is a huge resource and the librarians are eager to help you with this (or any other) class!

> **Note**
>
> Please note that for anything denoted as an E-Book at the library, there are limits placed on how many people can view the source. Instead, you can download sections (or whole books) for a period of time directly from the library.

6.e. Student Hours

Student hours are **drop-in hours.** These hours are for *you* to meet with me about anything! They are times on my calendar with *nothing else* on them.

I encourage you to add student hours to your weekly calendar. Below is an incomplete list of great reasons to come to student hours:

- You haven't made it to student hours yet
- You want to see where my office is
- You have a question on a reading, assignment, or activity
- You want to share about the course, your time at Smith, or about yourself
- You read something about data visualization, computer science, statistics, or math that you have questions about
- You are thinking about what you want to do after Smith
- You heard there's a robot in my office
- You are wandering campus and want somewhere to work for a minute

This semester, student hours are Mondays from 1:30pm to 3:30pm, Fridays from 9am to 10am, and by appointment.

Due to the pandemic, student hours and appointments will happen:

- in-person in my office or research space
- in-person, outdoors
- on zoom

6.e.i. *Appointments:*

You are welcome and encouraged to make individual appointments with me. To make an appointment, please check my appointments calendar (also linked on our Moodle site) for appointment slots that are set aside each week. If those times do not work for you, please slack me with at least two possible meeting times that work for you.

Please note that due to the pandemic, most appointments will be on zoom. If you prefer to meet in person, please do not hesitate to let me know.

6.e.ii. *Instructor information:*

The instructor for this course is Katherine Kinnaird. My office is room 208 in McConnell Hall. My email is kkinnaird@smith.edu. My student hours are Mondays from 1:30pm to 3:30pm, Fridays from 9am to 10am, and by appointment.

> **Note**
>
> To book an appointment with me, go to my appointment calendar

7. Academic Integrity

Nothing is held more dear to a researcher than their integrity. To researchers, the most precious commodity is our ideas. While some researchers build tangible items that can be protected by patents, many researchers' products are ideas and theories, which are less likely to be protected under things like patents. Instead, it is academic integrity that governs our research communities, protecting our ideas from being stolen. The system of academic integrity relies on researchers trusting each other to be honest and to act with integrity.

7.a. Honor Code

Being at a school with an Honor Code, like Smith, is a special privilege. The Honor Code goes beyond inviting students to act with integrity, instead it welcomes students as equal participants of the learning community, imbuing students with the same level of trust that we extend to our collaborators and colleagues. The Smith College Honor Code, established in 1944, and as stated in the Student Handbook, says:

> Smith College expects all students to be honest and committed to the principles of academic and intellectual integrity in their preparation and submission of course work and examinations. Students and faculty at Smith are part of an academic community defined by its commitment to scholarship, which depends on scrupulous and attentive acknowledgement of all sources of information, and honest and respectful use of college resources.

This trust bestowed to students in the Honor Code is the same trust that exists in and among researchers within a research community, the same trust that exists between me and my collaborators. Simply put, at Smith, under the Honor Code, I trust that you will each act with integrity, citing sources when you celebrate others' ideas and noting who you work with when collaborating.

This trust manifests in how assignments are created, in how work is completed, and in how we resolve instances where the trust has been broken. At an Honor Code school, assignments are created knowing that while there are resources online that can offer complete solutions, you are trusted to not seek out such complete resources and that if you do stumble on a solution guide, you will not use it. Under an Honor Code, students are expected to keep careful notes about the resources that they consult and the people that they collaborate with. We get to assume that the work handed in by a student is the creation of that student. Lastly, when there are violations of the Honor Code, the resolution is determined by a committee established by the community and trusted to seek restoration of the whole community's trust through education and action.

I regard the Honor Code with deep and profound respect. Being an educator at an Honor Code school means we begin from a place of trust without any underlying suspicion of our students. Simply put, we - instructors and students - work from the assumption that all are acting in good faith and with the utmost integrity. This is an assumption that cannot be made at a school without an Honor code, and it is why it is a privilege to be both an instructor and a student at Smith under the Honor Code.

7.b. Honor Code Practicalities

The work you submit should be your own and created by you, unless explicitly listed as a group assignment. With the exception of the starred problems, I do encourage you to ask for help from your peers or myself when you have questions; however, copying is never allowed. The line between copying and helping is subtle. Below are a few guidelines:

- On *individual* assignments, do not share nor give your work with other students; instead, offer to discuss the big ideas of the task at hand.
- For *group* assignments, do not submit work without the consent of all students, nor should you share your group's work with other students. As with individual work, discussing the big ideas is encouraged.
- Do not look at someone else's work (including online solutions); instead, ask if you could talk with them about your ideas and share where you are getting stuck.
- Acknowledge those you talk or work with at the top of every assignment. I will not dock points for getting acknowledged help from others. If you generate a solution or an argument with someone, in addition to acknowledging that person, recreate the solution or argument on your own in your own words.

If you think you may have crossed the line between helping and copying, please talk to me. Do not let me discover that the line was crossed. If a violation of the Honor Code is suspected, the student will be informed and will be given the opportunity to meet with the instructor. As recommended by the Academic Honor Board and in keeping with Smith tradition, the student will be given time to self-report, and after such time, the suspected violation will be reported to the Academic Honor Board by the instructor.

If you are unsure about how Smith's Honor Code applies to our course, please consult with me or a class dean. We are happy to discuss the Honor Code with you.

7.c. Creation vs. Curation

In my role as your instructor, I am **curating** resources for you that I feel are best for the teaching and learning of statistics. In this process of curation, I will give credit to those whose materials I have used and I will only use materials as allowable by copyright and fair use.

In this course, you will be asked to **create** solutions to problem sets, lab reports, and a final project. In these acts of creation, you are demonstrating you most current understanding of the concepts in our course. This course will give you lots of chances to demonstrate what you know and lots of flexibility to repeatedly try again.

I am aware that there is a tension between the curation that I am modeling and the creating that is being asked of you. My goal is to provide the best resources to you, which requires reading and sorting through many sources (curation). Your goal is to learn as much statistics as deeply as possible, which requires a lot of practice, trying out statistical ideas, reflecting on your attempts, and trying again (creation).

8. Assessment

This course will use a version of specifications-based grading. In this course, there are 30 medals to collect: 15 Statistics Medals from the knowledge checks and 15 Habit Medals in other course work. The medals breakdown as follows

- 15 Statistics Medals: Knowledge Checks (with each medal tied to a particular skill)
- 3 Habit Medals: Engagement & Class Preparation
- 4 Habit Medals: Group Problem Sets (GPsets)
- 2 Habit Medals: Labs
- 4 Habit Medals: Group Project
- 2 Habit Medals: Starred Problems

8.a. Knowledge Checks

There are four knowledge checks throughout the term. These are just moments in time to share what you current understanding of the course material is. There are 15 medals, each representing core skills for the course. During each knowledge check, you can attempt as many or as few medals as you would like. Your goal, by the end of the semester, is to collect all 15 medals.

The first badge challenge will have the first 5 medals available, the second will have the first 10, and the third and fourth knowledge check will have all 15 medals. Put another way:

- Medals 1-5 will appear on knowledge checks 1-4
- Medals 6-10 will appear on knowledge checks 2-4
- Medals 11-15 will appear on knowledge checks 3 and 4.

You do not need to (nor should you) attempt all medals on each knowledge check, but you do need to collect all 15 medals by the end of the term.

8.a.i. *Levels for Statistics Medals on Knowledge Checks:*

There are four levels for each statistics medal:

- G – Gold: Deep understanding of the material; could talk about this concept with someone unfamiliar with statistics
- S – Silver: Surface understanding of the material; has facility with concept in statistical settings, but has not yet translated understanding to broader context; can articulate the formula but not yet explain what it means
- B – Bronze: Making progress on understanding material, but there are gaps to be filled in; would benefit from studying this material again
- X – cannot be assessed: Attempt missing and/or there are so many facts shared, it is not clear if the statistician understands what is and what is not relevant.

8.a.ii. *Achieving Statistics Medals on Knowledge Checks:*

The 15 medals associated to the knowledge checks are tied to the 15 statistics medals. The most recent level for each medal is the recorded level. This means that if you earn a gold level for medal 1 on the first knowledge check, then you never have to attempt this medal again. But, if you earn silver for medal 4 on the first challenge and then a bronze for medal 4 on the second challenge, your current level for medal 4 is bronze.

> **Note**
>
> You do not need to attempt all medals on each knowledge check. This means that you get to pick and choose which skills you are ready for on each knowledge check.

For example, you may elect to just study medals 1-4 for the first knowledge check, then 5-8 for the second one, 9-12 for the third, and then 13-15 for the final one (plus perhaps a few that you want to try again on the final).

Alternatively, you may elect to study all 5 medals for the first knowledge check to see how the system works, and then study 6-9 plus one from the first knowledge check that didn't go as well. Then for the third one, you might again study 4 new ones (say 10-13) plus one or two that you want to "polish" from the first nine medals that you studied for the first two knowledge checks.

> **Warning**
>
> You need to collect all 15 medals by the end of the term. This means that you will need to start to think a bit more strategically as you prepare for each knowledge check

> **Note**
>
> You **only** need to take the final knowledge check if you *want* to "upgrade" any medal levels.

8.b. Habit Medals

The 15 habit medals are spread over the activities in this course that help develop your habits as a statistician. Like the statistics medals, there are four levels for each medal: Gold, Silver, Bronze, X (can't be assessed)

8.b.i. *Engagement & Class Preparation: 3 Habit Medals:*

Two of these habit medals will be earned with student input. In the sixth week of the course and during the last week of classes, you will be asked to complete a self-evaluation of your class engagement in context of the class engagement list. Your self-evaluations will be a critical part of determining the levels associated to these three medals, but not the only determining factor.

The third habit medal for class engagement is tied to your completion of the two individual writing assignments. If you complete both in good faith, you earn a Gold. If you only complete one in good faith, then you earn a Silver.

8.b.ii. *Group Problem Sets (GPsets): 4 Medals:*

There are 31 GPsets, each worth 30 points, for a total of 930 points. A **total** score of 744 points is equivalent to 4 gold levels. This means that you can either elect to hand in 25 of the 31 GPsets (effectively dropping 6 of them) or you can average 24 points on each of the 31 GPsets.

> **Note**
>
> You do not need my permission to not hand in all or part of an assignment. You might elect not to hand in (all or part of) an assignment for any number of reasons: it could be because you are ill, you forgot about it, or you had too much work for your other courses.

> **Warning**
>
> Note that pre-planned "dropping" of assignments should be used sparingly because we can not predict when life will happen (ie. illness, etc).

8.b.iii. *Labs: 2 Habit Medals:*

Updated December 17, 2021

There are 10 labs each with a pre-lab. All labs and pre-labs are each graded on completion, where **completion** means a good faith attempt. To earn a gold for the pre-lab medal, you need to complete 8 (of the 10) pre-labs with a good faith attempt. To earn a gold for the lab medal, you need to complete 7 (of the 10) labs with a good faith attempt. Like the GPsets, this means that you elect to not hand up to 3 of the labs and 3 of the pre-labs and still earn full credit for these two medals. For a silver for either the lab or pre-lab medal, you need to complete 5 (of 10) in good faith.

You do not need my permission to not hand in a lab or pre-lab. You might elect not to hand in a lab or pre-lab for any number of reasons: it could be because you are ill, you forgot about it, or you had too much work for your other courses.

8.b.iv. *Group Project: 4 Habit Medals:*

The habit medals associated to the group project will be tied to a rubric. That rubric will be introduced with the project.

8.b.v. *Starred Problems: 2 Habit Medals:*

Updated December 17, 2021

There are 10 starred problems, each with two submission steps: one on Moodle and one on Gradescope. These problems are based on completion (ie. you get the credit or not) based on both your solution and the content of your reflection. This means that you can get the problem totally wrong and still get full credit if you take the time to honestly reflect on your progress.

To earn two gold habit medals for the starred problems, you need to complete 7 (of the 10) starred problems (with both submission steps) in good faith. Like the GPsets and labs, this means that you elect to not hand up to 3 of the starred problems to earn full credit for these two medals. For a silver for both medals, you need to complete 5 (of 10) in good faith.

You do not need my permission to not hand in a starred problem. You might elect not to hand in a starred problem for any number of reasons: it could be because you are ill, you forgot about it, or you had too much work for your other courses.

8.c. Flexibility Systems

This course has a number of flexbility systems built in:

- Notice that for GPsets, Labs, and Starred Problems, you only need to complete about 80% of the work in each division to earn a top score for that division.
- For the statistics medals on the knowledge checks, you have between 2 and 4 attempts for each medal. This means that you have time to iteratively study, attempt, reflect, and reattempt for each medal.

8.c.i. *Latework:*

All work is due at 10am on the date indicated on the Detailed Course Schedule. This is a great moment to note the due dates of all assignments in the system (ie. planner, calendar, etc) that you use to keep track of your work. (If you want to really stretch here, do this for all major assignments in all your courses.)

In general, work should not be late. Later course content builds on previous content, and if one is spending time on an assignment that is passed due, then there is less time for the next assignment. Additionally the grading for this course has a number of flexibility systems in place, and I encourage that you take advantage of them as needed.

This all being said, there's a pandemic, and life is not normal. If you find yourself in need of a bit more time, send me a brief slack message at least a day before the work is due. In your message include 1) what flexibility systems you've already used (see above), 2) why it is important to your learning that you get an extension, 3) propose a new due date for yourself, and 4) why the proposed extension will not hinder your ability to get the next assignment done on time. Your message should be *no more than seven sentences long.*

8.c.ii. *Missing a knowledge check:*

The self-schedule knowledge check windows are listed on the Detailed Course Schedule. That being said, things do come up and you may be in a position where you need to miss a knowledge check. Under this grading system, you are working to demonstrate you most complete understanding of the course material by the end of the term. So you can miss a knowledge check for any reason knowing that you can attempt the medals from the missed challenge on a later knowledge check.

Warning

Missing two or more knowledge checks is not advisable.

This being said, if something comes up, I would rather you reach out and check in, than not reach out.

8.d. Turning Medals into Letter Grades

Traditional letter grades will be assigned using the below listed minimum medal levels:

Letter Grade	Number of Gold	Number of Silver	Number of Bronze	X
A	21	6	3	0
B	15	9	5	1
C	9	8	9	4
D	3	7	12	8

Note

Higher levels can count for lower level totals, but not vice versa. For example, a silver level can count towards your bronze total but a silver level cannot count towards your gold total.

At the instructor's discretion, these grade cutoffs may be lowered at the end of the semester, but they will never be raised. Cut-offs not stated here for the pluses and minuses will be determined at the end of the semester.